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Distribution of non-*aureus* staphylococci from quarter milk, teat apices, and rectal feces of dairy cows, and their virulence potential

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ABSTRACT

Non-*aureus* staphylococci (NAS) are predominantly isolated from bovine milk samples of quarters suffering from subclinical mastitis. They are also abundantly present on dairy cows' teat apices and can be recovered from bovine fecal samples, as recently described. Differences in ecology, epidemiology, effect on udder health, and virulence or protective traits have been reported among the species within this group. The objectives of this study were (1) to describe the species-specific distribution of NAS in 3 bovine-associated habitats, namely quarter milk, teat apices, and rectal feces, and (2) to evaluate the virulence potential of NAS by comparing their distribution in contrasting milk sample strata and the presence of selected virulence genes. A cross-sectional, systematic sampling procedure was followed in 8 dairy herds that participated in the local Dairy Herd Improvement program in Flanders, Belgium. Quarter milk samples (n = 573) were collected from 144 lactating cows in 8 herds. In 5 of the 8 herds, teat apex swabs (n = 192) were taken from 15 lactating cows, before and after milking, and from 18 dry cows. In the same 5 herds, rectal feces were sampled from 80 lactating cows (n = 80), taking into account that a cow could only serve as the source of one type of sample. In addition, milk samples of all clinical mastitis cases were continuously collected during the 1-yr study period from March 2017 to March 2018 in the 8 herds. In total, 1,676 *Staphylococcus* isolates were phenotypically identified and subjected to MALDI-TOF mass spectrometry. Thirty-three, 98, and 28% of all quarter milk, teat apex, and rectal fecal samples were NAS-positive, respectively, reaffirming the presence of NAS in rectal feces. The overall predominant species in the 3 habitats combined were *Staphylococcus haemolyticus*, *Staphylococcus chromogenes*, and *Staphylococcus hominis*. Four, 16, and 12% of the healthy quarters (quarter milk somatic cell count ≤50,000 cells/mL of milk), quarters with subclinical mastitis (quarter milk somatic cell count >50,000 cells/mL of milk), and quarters with clinical mastitis, respectively, were NAS-positive, suggesting that the potential to cause (mild) clinical mastitis is present among NAS. This was substantiated by comparing the presence of virulence genes of NAS isolates originating from contrasting milk sample strata (healthy quarters and quarters with clinical mastitis).
Key words: mastitis, non-*aureus* staphylococci, rectal feces, teat apex

INTRODUCTION

Non-*aureus* staphylococci are the most frequently isolated bacteria causing IMI in dairy cows worldwide (Sampimon et al., 2009b; Schukken et al., 2009; Condas et al., 2017b). They also colonize teat apices of both lactating and dry cows (Braem et al., 2013; De Visscher et al., 2016b; Adkins et al., 2018a) and can be present in feces of dairy cows (Wuytack et al., 2019). Whether milk, teat apices, and feces have comparable NAS distributions remains unclear.

The relevance of NAS IMI for bovine udder health and milk yield is still not entirely clear. Non-*aureus* staphylococci, as a group, are minor pathogens, but species- and strain-specific research revealed variation in virulence among NAS species (Vanderhaeghen et al., 2014). An increase in SCC comparable to a *Staphylococcus aureus* IMI (Supré et al., 2011; Fry et al., 2014; Valckenier et al., 2020) and the ability to cause persistent IMI (Rajala-Schultz et al., 2009; Mørk et al., 2012; Valckenier et al., 2020) or even mild clinical mastitis (Taponen et al., 2006; Simojoki et al., 2009; Verbeke et al., 2014) have been reported. At the same time, protective effects against other major mastitis pathogens of NAS IMI or NAS teat apex colonization have been described (De Vliegher et al., 2004; Piepers et al., 2009)

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and several authors describe little to no negative effects on udder health (Matthews et al., 1991; Reyher et al., 2012) and milk yield (Pearson et al., 2013; Piepers et al., 2013; Valckenier et al., 2019). Additional research into the virulence potential of NAS (e.g., by comparing their presence in milk samples from healthy quarters and from quarters with clinical mastitis and by comparing the frequency of virulence genes in NAS isolates originating from these 2 opposite strata) would be helpful in that respect. Methicillin-resistance and virulence determinants may be combined on the staphylococcal chromosomal cassette *mec* (**SCC*mec***) element without reducing the competitive fitness of the *Staphylococcus* spp. (Diep et al., 2008; Queck et al., 2009; Pozzi et al., 2012). The presence of methicillin-resistance genes was, therefore, determined as well.

The main objective of this study was to broaden our knowledge of the species-specific NAS presence and distribution in 3 bovine-associated habitats, namely quarter milk, teat apices, and rectal feces. Second, the (species-specific) virulence potential of NAS was evaluated by comparing their presence in milk samples originating from quarters with a low or elevated quarter milk SCC or with signs of clinical mastitis, and by determining the presence of selected virulence and methicillin-resistance genes in NAS originating from contrasting milk sample strata (low quarter milk SCC and clinical mastitis).

MATERIALS AND METHODS

Herds, Cows, and Samples

Eight commercial dairy herds (**H1**–**H8**) were enrolled in this observational study, conducted between March 2017 and March 2018. All herds participated in the local DHI program in Flanders, Belgium, with an interval of 4 to 6 wk between 2 test-days (CRV, Arnhem, the Netherlands). The herds had an average to good udder health management based on the implementation of the mastitis prevention and control program of the National Mastitis Council (**NMC**; National Mastitis Council, 2010). During the study, the average number of lactating cows per herd was 118 dairy cows (ranging from 50 to 237 cows) with a yearly average milk production per cow of 9,304 kg of milk (ranging from 8,646 to 12,059 kg of milk) and a geometric mean of the monthly bulk milk SCC of 121,000 cells/mL of milk (ranging from 77,000 to 188,000 cells/mL of milk). The teat disinfection products used before milking were either none (H2, H3, H5, and H7) or foam based on low concentration of lactic acid (H1, H4, H6, and H8; Keno pure, CID LINES, Ieper, Belgium, https://www.cidlines.com/en-INT/kenotmpure) and after the milking process were either based on iodine (H1, H2, H4, H6, and H8; IoShield, Ecolab, St. Paul, MN, https://en-uk.ecolab.com/offerings/post-milking-udder-care/io-shield; VelouCid, Ecolab, https://en-uk.ecolab.com/offerings/post-milking-udder-care/veloucid-d; Blockade, DeLaval, Cardiff, UK, https://www.delaval.com/en-gb/our-solutions/milking/udder-health--hygiene/teat-dips/) or chloordioxide (H3, H5, and H7; Dippingold Comfort, Heemskerk, Diessen, the Netherlands, https://dailymed.nlm.nih.gov/dailymed/fda/fdaDrugXsl.cfm?setid=f93303bd-451c-4fd3-99d7-370e2eb444e9&type=display), following the manufacturers' guidelines.

A single cross-sectional sampling of each herd was performed during which quarter milk samples from approximately 25% (maximum 80 milk samples per herd, Table 1, Figure 1) of the lactating animals per herd were collected via systematic random sampling for bacteriological culturing based on the guidelines of the NMC (Hogan et al., 1999). As well, milk samples of all quarters with clinical signs of mastitis were collected secundum artem by the producers during the entire study period. Based on the severity of the disease signs, clinical mastitis cases were categorized as mild, moderate, or severe, following the scoring system described by Kalmus et al. (2013). The quarter milk SCC (**qSCC**) was determined from all milk samples except from quarters with signs of clinical mastitis, using a Direct Cell Counter (DeLaval, Tumba, Sweden) and used to stratify the nonclinical samples in low qSCC (≤50,000 cells/mL) milk samples (~healthy quarters) and elevated qSCC (>50,000 cells/mL) milk samples (considered as quarters with subclinical mastitis; Leitner et al., 2003; Barbano et al., 2006; Potter et al., 2018).

In 5 of the 8 herds (H1, H2, H3, H4, and H5), swabs from teat apices (Table 1, Figure 1) were collected at the same time the milk samples were collected. The 4 teat apices of both lactating and dry cows (total number of animals equaled approximately 10% of the total number of lactating cows per herd) were sampled in a systematic random way [excluding cows already sampled for milk or feces (see further)]. Approximately one-third of the sampled teat apices originated from lactating cows and two-thirds from dry cows. If the number of dry cows was insufficient at the moment of the cross-sectional sampling, extra teat apices of lactating cows were sampled. The teat apices of the lactating cows were sampled twice, once before and once after milking at sampling day. Samples were collected with a dry cotton swab (Copan, Novolab, Belgium) as described by De Visscher et al. (2014). The teat disinfection products used before milking are described above.

In the same 5 herds, rectal feces samples (Table 1, Figure 1) were collected from approximately 25% of the

lactating cows per herd (excluding cows sampled for milk or teat apices) via systematic random sampling, using double gloves reflecting a guarded sampling technique (Wuytack et al., 2019).

All samples were transported under cooled conditions (4°C) to the Mastitis and Milk Quality Research Laboratory (Faculty of Veterinary Medicine, Ghent University, Merelbeke, Belgium).

Bacterial Isolation and Identification

Quarter milk samples were plated with a 0.01-mL loop on esculin blood agar (Oxoid, Aalst, Belgium), MacConkey agar (Oxoid), and mannitol salt agar (**MSA**, Chapman medium, Oxoid), a semi-selective agar for staphylococci (De Visscher et al., 2013). Swabs collected from the teat apices were streaked on MSA to prevent overgrowth of other bacteria, which is frequently observed in samples from teat apices (Lugsomya et al., 2012; De Visscher et al., 2013). As described by Wuytack et al. (2019), rectal feces (1 g) were first diluted in 9 mL of 0.9% physiological salt solution (9 g of NaCl/L, Eurovet, Bladel, the Netherlands) and then plated with a 0.01-mL loop on MSA. All plates were incubated at 37°C and examined after 24 and 48 h. All phenotypically different colony types (with ≥1 colony) per plate were counted and subcultured on blood agar (Columbia agar with 5% sheep blood, Oxoid) and phenotypical bacteriological identification was performed following the NMC procedure guidelines (Hogan et al., 1999). The isolates identified as *Staphylococcus* spp. were stored at −80°C until further analysis.

All isolates phenotypically identified as *Staphylococcus* spp. were submitted to MALDI-TOF with a custom reference-spectra expanded database for bovine-associated NAS (Cameron et al., 2018). The MALDI-TOF assay was performed using the direct transfer method with a cut-off value of ≥1.700 for NAS isolates (Cameron et al., 2017). In addition, the MALDI library was updated as described by Cameron et al. (2017) with the following missing species: *Staphylococcus jettensis*, *Staphylococcus lentus*, *Staphylococcus rostri*, and *Staphylococcus saprophyticus* and some additional bovine isolates belonging to *Staphylococcus chromogenes*, *Staphylococcus epidermidis*, *Staphylococcus equorum*, *Staphylococcus fleurettii*, *Staphylococcus haemolyticus*, *Staphylococcus hyicus*, and *Staphylococcus simulans.*

Descriptive Analysis

To picture the overall distribution of NAS in the different habitats, all isolates originating from quarter milk (with or without an elevated qSCC or clinical signs), teat apices, and rectal feces were included for the descriptive analysis. Quarter milk samples harboring more than 2 different microorganisms on either MSA, esculin blood agar, and MacConkey agar were considered contaminated and excluded from the analysis. For this analysis, NAS were considered as one group, meaning that samples could harbor isolates belonging to more than 2 different NAS species.

The (species-specific) virulence potential of NAS was evaluated based on their distribution among the contrasting milk sample strata (healthy quarters with a qSCC ≤50,000 cells/mL and quarters with clinical signs) and by comparing the presence of 15 virulence genes in the NAS isolates (n = 60; 17 from quarters with a low qSCC and 43 from clinical quarters). To do so, NAS isolates from quarter milk samples harboring NAS only were used, meaning that quarter milk sam-

Table 1. The number of samples collected per dairy herd and per habitat (quarter milk, teat apices, and rectal feces) during the cross-sectional and continuous sampling period of 1 yr for clinical mastitis cases

	Herd (H)								
	H1 (44)[1]	H2 (94)	H3 (62)	H4 (64)	H5 (67)	H6[2] (187)	H7[2] (203)	H8[2] (93)	
Habitat	Number of samples								Total
Quarter milk	59	115	64	98	100	116	159	92	803
Healthy quarters (geometric mean 20,000 cells/mL)	16	37	37	23	52	41	47	59	312
Subclinical signs (geometric mean 152,000 cells/mL)	32	42	27	51	16	39	33	21	261
Clinical signs	11	36	—	24	32	36	79	12	230
Teat apices	32	44	40	32	44	—[2]	—	—	192
Lactating cows	16	24	32	16	32	—	—	—	120
Before milking	8	12	16	8	16	—	—	—	60
After milking	8	12	16	8	16	—	—	—	60
Dry cows	16	20	8	16	12	—	—	—	72
Rectal feces	11	20	16	16	17	—	—	—	80

[1]The number of lactating cows present per herd during the cross-sectional sampling is shown in parentheses.

[2]In herds 6, 7, and 8, only quarter milk samples were collected.

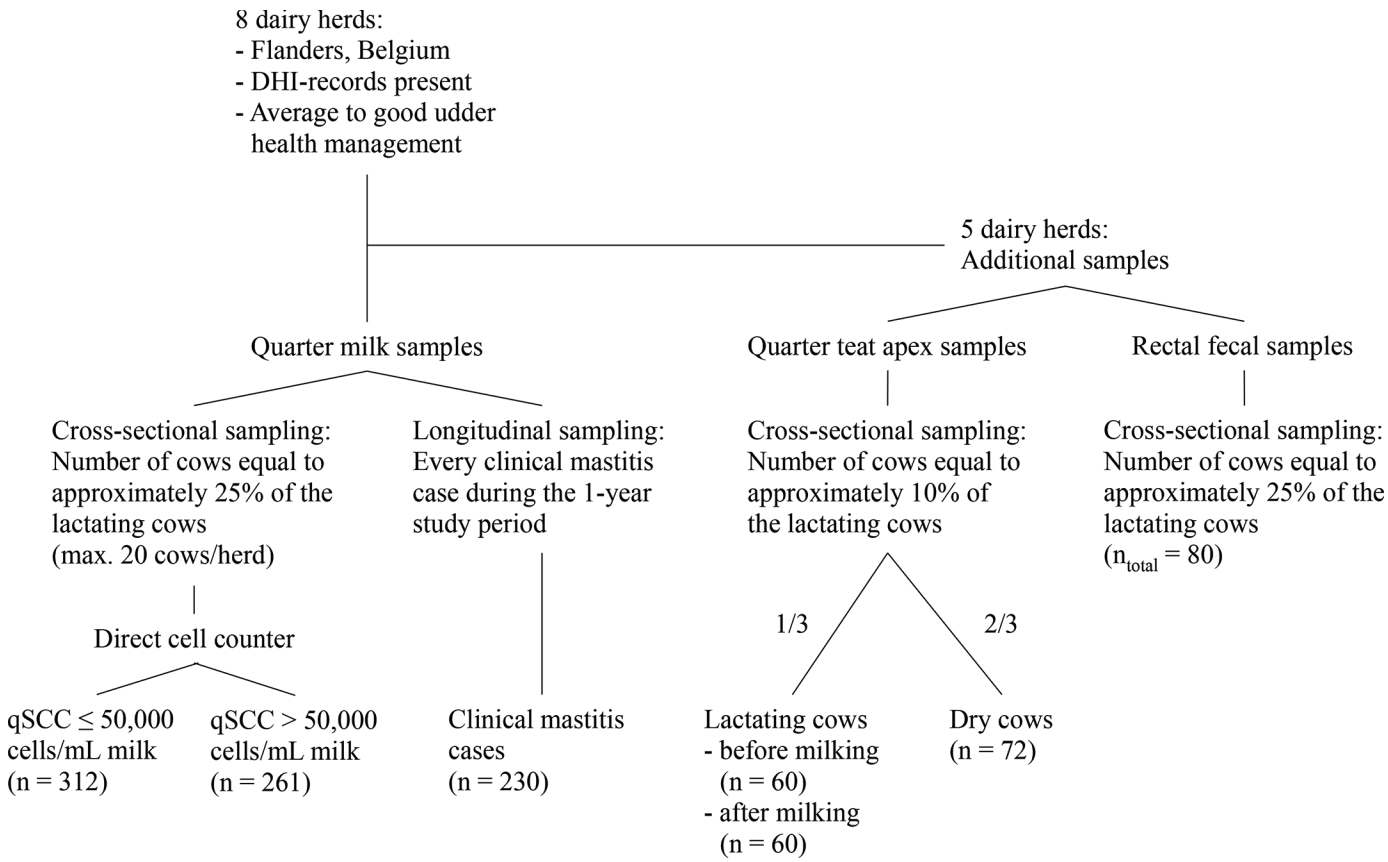

Figure 1. Overview of the study design. qSCC = quarter milk somatic cell count; max. = maximum.

ples harboring bacteria other than NAS as evidenced on MSA, esculin blood agar, or MacConkey agar (or a combination of these) were excluded for this analysis.

Statistical Analysis

Using ANOVA, differences between NAS as a group and major pathogens in effect on qSCC were analyzed statistically as well as the effect of the most prevalent NAS species (*S. chromogenes* and *S. haemolyticus*; IBM SPSS Statistics version 25, Armonk, NY). Only milk samples harboring microorganisms in pure culture were included for this analysis. A natural log-transformation of qSCC was performed to normalize the date. Bonferroni-adjusted *P*-values were generated to adjust for multiple comparisons.

PCR Assay

The DNA was extracted from the 60 NAS isolates originating from the 2 contrasting milk sample strata with the use of the commercial DNeasy Blood and Tissue Kit (Qiagen, Hilden, Germany) based on the manufacturer's instructions. One isolate originating from a healthy quarter from herd 5 was, however, not revivable from the −80°C.

Kit reagents without sample were used as negative control. The virulence and methicillin-resistance associated genes tested with the use of specific PCR amplifications were *agrA* (accessory gene regulator protein), *bap* (biofilm-associated protein), *cap5H* and *8H* (CP5 and CP8 synthesis enzyme), *clfA* (clumping factor), *coa* (coagulase), *eno* (laminin binding protein), *fnbpA* and *B* (fibronectin-binding protein A and B), *hla* (α-hemolysin), *hlb* (β-hemolysin), *ica* (polysaccharide intercellular adhesion), *mecA* and *mecC* (methicillin resistance), *nuc* (thermonuclease), *ssl7* (staphylococcal superantigen-like protein), and *vwb* (von-Willebrand factor binding protein). The PCR reactions for the virulence and methicillin-resistance genes were performed according to Wang et al. (2016) with the following modification: the PCR reaction mixture (25 μL) contained 2X ALLin HS Red Taq Mastermix (Labconsult, Brussels, Belgium) instead of 2X Taq PCR MasterMix. The primer sequences, annealing temperatures, and positive controls used for the detection of the virulence

and methicillin-resistance genes are shown in Table 2. In each PCR run, a positive and negative control (without DNA template) were included. The PCR products were analyzed by agarose gel electrophoresis (1.5% wt/vol), previously stained with ethidium bromide (Sigma-Aldrich, Overijse, Belgium).

RESULTS

Overall Distribution of NAS

Quarter Milk, Teat Apices, and Rectal Feces. In total, 1,676 isolates were phenotypically identified as *Staphylococcus* spp. and submitted to the MALDI-TOF MS. In sum, 166 isolates had either no identification (51 isolates, score <1.700) or were identified as *Aerococcus* spp., *Arthrobacter* spp., *Corynebacterium* spp., *Enterococcus* spp., *Kocuria* spp., or *Micrococcus* spp. (score ≥1.700), resulting in 1,510 *Staphylococcus* isolates (score ≥1.700). The NAS distribution in the 3 examined habitats, quarter milk, teat apices, and rectal feces, is tabulated in Table 3. For 9 species, isolates were present in all 3 habitats in the same herd, namely *Staphylococcus arlettae*, *Staphylococcus auricularis*, *S. chromogenes*, *Staphylococcus cohnii*, *Staphylococcus devriesei*, *Staphylococcus equorum*, *S. haemolyticus*, *Staphylococcus hominis*, and *Staphylococcus vitulinus*. *Staphylococcus haemolyticus* and *S. hominis* were even represented in all habitats in 2 (herds 1 and 5), and 4 (herds 1, 2, 3, and 4) herds, respectively. Other species were found in only 1 habitat, namely *Staphylococcus agnetis*, *Staphylococcus kloosii*, and *Staphylococcus pettenkoferi* on teat apices and *Staphylococcus hyicus*, *Staphylococcus schleiferi*, and *Staphylococcus warneri* in quarter milk.

Quarter Milk. In total, 803 quarter milk samples originating from 749 quarters of 301 cows, were collected, of which 39%, 32%, and 29% were originating from healthy quarters, quarters with subclinical mastitis, and quarters with signs of clinical mastitis, respectively (Table 1). Table 4 gives an overview of the NAS-positive samples per herd for the overall distribution of NAS. Allowing for pathogens other than NAS to be present, 33% of the quarter milk samples were NAS-positive (range 18 to 50% per herd), harboring 22 NAS species (range 5 to 15 species per herd). The 5 most prevalent species were *S. chromogenes* (10% positive quarter milk samples), *S. haemolyticus* (9%), *S. equorum* (7%), *S. hominis* (5%), and *S. cohnii* (3%). *Staphylococcus chromogenes*, *S. haemolyticus*, and *S. hominis* were present in quarter milk on all farms.

Teat Apices. In total, 192 teat apex samples from 33 cows in 5 herds were sampled (Table 1). This included samples from 15 lactating cows sampled before and after the milking process, as well as samples from 18 dry cows. All teat apex samples harbored NAS species except for 8 (4.2%), which were all taken after the milking process. Up to 8 different NAS species per teat apex sample could be identified, resulting in 830 NAS isolates, belonging to 22 different NAS species (Table 4). The 5 most frequently isolated NAS species were *S. haemolyticus* (70% positive teat apex samples), *S. equorum* (33%), *S. chromogenes* (23%), *S. cohnii* (19%), and *S. devriesei* (15%; Table 3). *Staphylococcus cohnii* was present in only 2 herds, whereas *S. devriesei* was present in 4 of the 5 sampled herds. *Staphylococcus auricularis*, *S. chromogenes*, *S. equorum*, and *S. haemolyticus* were isolated from teat apices on every farm.

Table 5 outlines the distribution of NAS species from the teat apices of the lactating cows (n = 15), swabbed before and after milking, and dry cows for the different herds. *Staphylococcus haemolyticus* was found on all sampled teat apices before milking in herd 1. *Staphylococcus chromogenes* and *S. haemolyticus* were present on all sampled teat apices from dry cows in herd 3. *Staphylococcus kloosii*, *S. lentus*, and *S. saprophyticus* were only picked up from teat apices of the lactating cows before the milking process, whereas *S. agnetis* and *S. epidermidis* were only found after milking. The following species could only be isolated from the teat apices of dry cows: *S. fleurettii*, *S. pettenkoferi*, *S. rostri*, and *S. simulans*. *Staphylococcus agnetis*, *S. kloosii*, *S. lentus*, and *S. saprophyticus* could not be retrieved from teat apices of dry cows.

Rectal Feces. Rectal fecal samples of 80 cows from 5 herds were collected (Table 1). Twenty-eight percent (ranging from 6% to 55% per herd) was NAS-positive, resulting in 33 different isolates belonging to 12 species (Table 4). Table 3 describes the distribution of NAS species for the different herds. All herds had at least one dairy cow that was shedding NAS. The most prevalent species were *S. hominis* (31% or the rectal fecal samples), *S. epidermidis* (15%), *S. cohnii* (12%), and *S. haemolyticus* (8%). Other NAS species present in rectal feces were *S. arlettae*, *S. auricularis*, *S. capitis*, *S. chromogenes*, *S. devriesei*, *S. equorum*, *S. lentus*, and *S. vitulinus*.

Virulence Potential of NAS in Contrasting Milk Strata

NAS Presence. During the cross-sectional sampling, 573 clinically healthy quarters belonging to 144 cows were sampled (Table 1). Three-quarters were blind. Three hundred twelve (54%) and 261 (46%) quarter milk samples had a qSCC ≤ or >50,000 cells/mL of milk, respectively. During the sampling period of 1 yr, 188 cows showed signs of clinical mastitis. Of those 188 cows, 28 cows had more than one clinical mastitis case

Table 2. Target genes, primers, and PCR conditions used for the amplification of virulence and methicillin-resistance genes

Target gene	Protein	Primer pair[1]	Primer sequence (5′–3′)	Temperature (°C)	Positive control	Reference
	Antimicrobial susceptibility					
mecA	Methicillin resistance	*mecA1*	AAA ATC GAT GGT AAA GGT TGG C	55	*Staphylococcus aureus* ATTC 33591	Murakami et al. (1991)
		mecA2	AGT TCT GCA GTA CCG GAT TTG C			
mecC	Methicillin resistance	*mecC*	TCA CCA GGT TCA AC(Y) CAA AA	55	*S. aureus* LGA 251	García-Álvarez et al. (2011)
		mecC	CCT GAA TC(W) GCT AAT AAT ATT TC			
	Exoenzymes					
coa	Coagulase		ATA GAG ATG CTG GTA CAG G	55	*S. aureus* ATCC 29213	Mahmoudi et al. (2017)
			GCT TCC GAT TGT TCG ATG C			
vwb	von Willebrand factor binding protein		AAT ATA CCA TGG TGG TTT CTG GGG AGA AGA AT	55	*S. aureus* ATCC 25923	Bjerketorp et al. (2002)
			TTT GCC ATT ATA TAC TTT ATT GAT			
nuc	Thermonuclease	*nuc*-F	GCG ATT GAT GGT GAT ACG GGT	57	*S. aureus* ATCC 25923	Wang et al. (2016)
		nuc-R	AGC CAA GCC TTG ACG AAC TAA AGC			
	Exotoxins					
hla	α-Hemolysin precursor	*hla*-F	GGT TTA GCC TGG CCT TC	57	*S. aureus* ATCC 25923	Wang et al. (2016)
		hla-R	CAT CAC GAA CTC GTT CG			
hlb	β-Hemolysin precursor	*hlb*-F	GCC AAA GCC GAA TCT AAG	58	*S. aureus* ATCC 33591	Wang et al. (2016)
		hlb-R	CGC ATA TAC ATC CCA TGG C			
ssl7	Staphylococcal superantigen-like protein	*ssl7*-F	GGT TTA TTC ATA GCG CAG TAT C	58	*S. aureus* ATCC 25923	Salasia et al. (2004)
		ssl7-R	CAA CGT TTC ATC GTT AAG CTG C			
	Regulatory proteins					
agr	Accessory gene regulator protein		ATG CAC ATG GTG CAC ATG CA	55	*S. aureus* ATCC 25923	Dufour et al. (2002)
			CAT AAT CAT GAC GGA ACT TGC TGC GCA			
	Surface factors					
bap	Biofilm-associated protein	*bapF-6m*	ACT TAY TRC CHT ATA TCG AAR TAG	57	*S. xyloxus* ATCC 29971	Cucarella et al. (2004)
		Sasp-7c	GCT GTT GAA GTT AAT ACT GTA CCT GC			
cap5H	CP5 synthesis enzyme	*cap5H*	ATG AGG ATA GCG ATT GAA AA	50	*S. aureus* NCTC 8325–4	Ote et al. (2011)
		cap5R	CGC TTC TTA ATC ACT TTT GC			
cap8H	CP8 synthesis enzyme	*cap8H*	ATC GAA GAA CAT ATC CAA GG	50	*S. aureus* ATCC 33591	Ote et al. (2011)
		cap8R	TTC ATC ACC AAT ACC TTT TA			
clfA	Clumping factor ClfA	*clfA*-F	GGC AAC GAA TCA AGC TAA TAC AC	58	*S. aureus* ATCC 25923	Wang et al. (2016)
		clfA-R	TTG TAC TAC CTA TGC CAG TTG TC			
eno	Laminin binding protein	eno-1	ACG TGC AGC AGC TGA CT	55	*S. aureus* ATCC 29213	Salaberry et al. (2015)
		eno-2	CAA CAG CAT CTT CAG TAC CTT C			
fnbpA	Fibronectin-binding protein A	*fnbpA*-F	GCG GAG ATC AAA GAC AA	50	*S. aureus* ATCC 25923	Wang et al. (2016)
		fnbpA-R	CAT CTA TAG CTG TGT GG			
fnbpB	Fibronectin-binding protein B	*fnbpB*-F	GGA GAA GGA ATT AAG GCG	50	*S. aureus* ATCC 33591	Wang et al. (2016)
		fnbpB-R	GCC GTC GCC TTG AGC GT			
ica	Polysaccharide intercellular adhesion	*icaFSe*	CTG TTT CAT GGA AAC TCC	57	*S. epidermidis* ATCC 35984	Rohde et al. (2007)
		icaRSe	TCG ATG CGA TTT GTT CAA ACA T			

[1]F = forward; R = reverse.

Table 3. Distribution of 1,238 phenotypically different non-*aureus* staphylococci (NAS) originating from quarter milk (QM), teat apices (TA), and rectal fecal (RF) samples with mixed culture in 8 commercial dairy herds

		Herd (H)																	
		H1		H2		H3		H4		H5		H6		H7		H8		Total	
Species	Habitat	N[1]	%[2]	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%
Staphylococcus agnetis	QM																		
	TA					1	3					—[3]	—	—	—	—	—	1	1
	RF											—	—	—	—	—	—		
Staphylococcus arlettae	QM			2	2									2	2			4	1
	TA	1	3	5	9	2	3			5	11	—	—	—	—	—	—	13	6
	RF			1	14							—	—	—	—	—	—	1	4
Staphylococcus auricularis	QM	1	2	1	1	1	4	2	3			2	2					7	1
	TA	8	13	3	7	4	10	23	44	3	7	—	—	—	—	—	—	41	15
	RF							1	20			—	—	—	—	—	—	1	4
Staphylococcus capitis	QM							1	1					3	2			4	1
	TA	3	9			1	3	7	19			—	—	—	—	—	—	11	5
	RF	1	14									—	—	—	—	—	—	1	4
Staphylococcus chromogenes	QM	11	14	12	9	5	15	11	9	4	7	29	18	4	4	7	7	83	10
	TA	8	25	1	2	29	50	3	9	16	30	—	—	—	—	—	—	57	23
	RF	1	14									—	—	—	—	—	—	1	4
Staphylococcus cohnii	QM			3	3			15	13					1	1	3	4	22	3
	TA			66	77			2	6			—	—	—	—	—	—	68	19
	RF	1	14	2	29							—	—	—	—	—	—	3	12
Staphylococcus devriesei	QM					1	4	9	10							1	1	11	2
	TA	1	3	9	16	1	3	45	59			—	—	—	—	—	—	56	15
	RF							2	40			—	—	—	—	—	—	2	8
Staphylococcus epidermidis	QM					1	4	8	8	1	2			8	6	3	4	21	3
	TA	4	9	17	27			1	3			—	—	—	—	—	—	22	8
	RF	1	14	2	14	2	50					—	—	—	—	—	—	5	15
Staphylococcus equorum	QM	4	8	2	2	4	7	5	5	4	7			31	19	6	6	56	7
	TA	60	72	8	18	10	25	2	6	26	48	—	—	—	—	—	—	106	33
	RF			1	14							—	—	—	—	—	—	1	4
Staphylococcus fleurettii	QM	1	2															1	1
	TA	4	13									—	—	—	—	—	—	4	2
	RF											—	—	—	—	—	—		
Staphylococcus haemolyticus	QM	9	16	7	6	18	30	18	19	3	6	17	10	3	2	3	4	78	9
	TA	54	81	80	77	77	83	38	66	27	45	—	—	—	—	—	—	276	70
	RF	2	14							1	33	—	—	—	—	—	—	3	8
Staphylococcus hominis	QM	5	8	1	1	4	15	13	13	3	6	6	5	1	1	5	4	38	5
	TA	17	22	19	32	7	15	6	16			—	—	—	—	—	—	49	17
	RF	1	14	8	57	1	25	2	40			—	—	—	—	—	—	12	31
Staphylococcus hyicus	QM											2	1					2	1
	TA											—	—	—	—	—	—		
	RF											—	—	—	—	—	—		
Staphylococcus kloosii	QM																		
	TA			2	2							—	—	—	—	—	—	2	1
	RF											—	—	—	—	—	—		
Staphylococcus lentus	QM							1	1					1	1			2	1
	TA	1	3									—	—	—	—	—	—	1	1
	RF							1	20			—	—	—	—	—	—	1	4

Continued

Table 3 (Continued). Distribution of 1,238 phenotypically different non-*aureus* staphylococci (NAS) originating from quarter milk (QM), teat apices (TA), and rectal fecal (RF) samples with mixed culture in 8 commercial dairy herds

		Herd (H)																	
		H1		H2		H3		H4		H5		H6		H7		H8		Total	
Species	Habitat	N[1]	%[2]	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%
Staphylococcus pettenkoferi	QM																		
	TA							3	9			—	—	—	—	—	—	3	2
	RF											—	—	—	—	—	—		
Staphylococcus rostri	QM													1	1			1	1
	TA									1	2	—	—	—	—	—	—	1	1
	RF											—	—	—	—	—	—		
Staphylococcus saprophyticus	QM			1	1			1	1					1	1			3	1
	TA					1	3					—	—	—	—	—	—	1	1
	RF											—	—	—	—	—	—		
Staphylococcus schleiferi	QM											1	1					1	1
	TA											—	—	—	—	—	—		
	RF											—	—	—	—	—	—		
Staphylococcus sciuri	QM	3	4	1	1									1	1			5	1
	TA	26	50	9	20			1	3			—	—	—	—	—	—	36	14
	RF											—	—	—	—	—	—		
Staphylococcus simulans	QM			1	1									1	1			2	1
	TA					1	3	7	13			—	—	—	—	—	—	8	3
	RF											—	—	—	—	—	—		
Staphylococcus succinus	QM			5	4									4	3			9	1
	TA	9	19	10	18					1	2	—	—	—	—	—	—	20	8
	RF											—	—	—	—	—	—		
Staphylococcus vitulinus	QM	3	6					2	1									5	1
	TA	7	16			1	3	1	3			—	—	—	—	—	—	9	4
	RF	2	14									—	—	—	—	—	—	2	4
Staphylococcus warneri	QM							1	1									1	1
	TA											—	—	—	—	—	—		
	RF											—	—	—	—	—	—		
Staphylococcus xylosus	QM	2	4	9	8			6	6					1	1	1	1	19	3
	TA	22	38	17	34			5	9	1	2	—	—	—	—	—	—	45	16
	RF											—	—	—	—	—	—		

[1]Number of phenotypically different (per sample) NAS isolates per habitat and herd.

[2]Percentage of samples positive for NAS per habitat and herd (samples can harbor more than one NAS species).

[3]In herds 6, 7, and 8, only quarter milk samples were collected.

in either the same or a different quarter, resulting in 230 clinical mastitis cases that were sampled. Twenty percent, 12%, and 8% of the quarter milk samples from healthy quarters, quarters with subclinical, or clinical signs were bacteriologically negative, respectively. Table 4 displays the percentage of samples only positive for NAS and the maximum number of NAS species per herd.

NAS Distribution. *Staphylococcus chromogenes*, *S. hominis*, and *S. xylosus* were present in all 3 types of quarter milk samples (i.e., healthy quarters, quarters with subclinical, or with clinical signs; Table 6). *Staphylococcus chromogenes* was the most prevalent species in healthy quarters or quarters with subclinical signs, and one of the 3 most commonly isolated species from quarters showing clinical signs. *Staphylococcus haemolyticus* was the second most and most frequently isolated NAS species from quarters with subclinical and quarters with clinical signs, respectively. In healthy quarters, *S. equorum*, *S. devriesei*, and *S. hominis* were equally present and thus the second most prevalent species. In quarters with subclinical signs, the third most prevalent species was *S. xylosus*. *Staphylococcus equorum* had the next highest prevalence in quarters with clinical signs. Based on the definitions according to Kalmus et al. (2013), milk samples from clinical mastitis cases harboring NAS only had score 1 (mild), 2 (moderate), or 3 (severe) in 70%, 15%, or 4% of the cases, respectively. Three milk samples originating from clinical mastitis harboring NAS only were not classified due to a lack of information on the severity score.

Quarter SCC. The NAS-positive milk samples (excluding those showing clinical signs, n = 57) had a geometric mean qSCC of 109,000 cells/mL of milk [interquartile range (**IQR**) from 52,000 to 210,000 cells/mL of milk]. Milk samples positive for a major mastitis pathogen only (*S. aureus*, *S. dysgalactiae*, *Serratia* spp., *Pasteurella multocida*, and *Trueperella pyogenes*, n = 6) had a geometric mean qSCC of 155,000 cells/mL of milk with a IQR from 60,000 cells/mL of milk to 830,000 cells/mL of milk, whereas samples yielding no growth (n = 93) had a geometric mean qSCC of 34,000 cells/mL of milk (IQR from 20,000 to 57,000 cells/mL of milk). Samples positive for *Corynebacterium* spp. (n = 60) had a geometric mean qSCC of 53,000 cells/mL (IQR from 24,000 to 101,000 cells/mL of milk). The NAS-positive milk samples (all species combined) had a significantly higher LnqSCC than the milk samples yielding no growth ($P = 0.000$) and *Corynebacterium* spp. ($P = 0.008$), yet the LnqSCC was not different from the major pathogen–positive milk samples. The geometric mean qSCC of samples positive for *S. chromogenes* (n = 20) or *S. haemolyticus* (n = 10) only were 156,000 cells/mL of milk (IQR from

Table 4. Samples positive for non-*aureus* staphylococci (NAS) allowing for other pathogens to be present in the 3 habitats quarter milk, teat apices, and rectal feces ("Overall distribution") and samples positive for NAS only in the 3 milk strata ("Virulence potential in milk strata") in commercial dairy herds

Item	$N_{totalsamples}$	$N_{NAS\ isolates}$	$Range_{NAS\ species/sample}$	% NAS-positive samples									Maximum number of different NAS species								
				Herd (H)									Herd (H)								Total
				H1	H2	H3	H4	H5	H6	H7	H8	Total	H1	H2	H3	H4	H5	H6	H7	H8	
Overall distribution																					
Habitat																					
Quarter milk	803	375	1–4	44	31	42	50	21	28	25	18	33	9	12	7	14	5	6	15	8	22
Teat apices	192	830	1–8	100	100	95	97	98	—[1]	—	—	98	15	13	12	14	8	—	—	—	22
Rectal feces	80	33	1–3	55	35	19	31	6	—	—	—	28	7	5	2	4	1	—	—	—	12
Total	1,075	1,238	1–8	63	48	57	58	40	29	28	28	44	16	15	13	17	10	6	15	8	25
Virulence potential in milk strata																					
Quarter milk stratum																					
Healthy quarter	312	17	1–2	0	0	0	22	2	2	4	8	4	0	0	0	4	1	1	3	4	8
Subclinical signs	261	74	1–4	13	14	4	31	0	26	6	19	16	3	2	1	7	0	3	1	2	8
Clinical signs	230	43	1–3	18	25	—	4	6	8	13	0	12	1	6	—	3	3	2	5	0	10
Total	803	134	1–4	10	13	2	22	3	12	9	10	10	4	6	1	10	4	3	8	5	15

[1]In herds 6, 7, and 8, only quarter milk samples were collected.

Table 5. Distribution of 833 phenotypically different non-*aureus* staphylococci (NAS) on teat apices originating from lactating cows (n = 15), sampled before and after milking, and from dry cows (n = 18) in 5 commercial dairy herds

		Herd (H)											
		H1		H2		H3		H4		H5		Total	
NAS species	Teat apex strata	N[1]	%[2]	N	%	N	%	N	%	N	%	N	%
Staphylococcus agnetis	Before milking												
	After milking					1	6					1	2
	Dry cows												
Staphylococcus arlettae	Before milking			3	17	2	6+			2	13	7	8
	After milking			2	17					1	6	3	5
	Dry cows	1	6							2	17	3	4
Staphylococcus auricularis	Before milking	2	13	1	8	2	13	7	50	1	6	13	15
	After milking					2	13	1	13	1	6	4	7
	Dry cows	6	19	2	10			15	56	1	8	24	21
Staphylococcus capitis	Before milking					1	6	5	50			6	8
	After milking	1	13					2	25			3	5
	Dry cows	2	13									2	3
Staphylococcus chromogenes	Before milking	1	13			9	50	1	13	7	38	18	27
	After milking	2	25			4	25			4	19	10	15
	Dry cows	5	31	1	5	16	100	2	13	5	33	29	28
Staphylococcus cohnii	Before milking			18	75							18	15
	After milking			16	75							16	15
	Dry cows			32	80			2	13			34	25
Staphylococcus devriesei	Before milking			5	33	1	6	15	75			21	18
	After milking							12	50			12	7
	Dry cows	1	6	4	15			18	56			23	18
Staphylococcus epidermidis	Before milking												
	After milking	3	25					1	13			4	5
	Dry cows	1	6	17	60							18	18
Staphylococcus equorum	Before milking	11	63	4	33	6	38	1	13	12	69	34	45
	After milking	8	50	2	17	4	25			9	31	23	25
	Dry cows	41	88	2	10			1	6	5	42	49	31
Staphylococcus fleurettii	Before milking												
	After milking												
	Dry cows	4	25									4	6
Staphylococcus haemolyticus	Before milking	24	100	20	92	33	94	8	50	8	44	93	75
	After milking	11	75	4	33	23	63	5	50	3	19	46	45
	Dry cows	19	75	56	95	21	100	25	81	16	83	137	86
Staphylococcus hominis	Before milking	13	63	5	25	4	19	3	25			25	22
	After milking	4	25	3	25	2	13	2	25			11	15
	Dry cows			11	40	1	13	1	6			13	14
Staphylococcus kloosii	Before milking			2	8							2	2
	After milking												
	Dry cows												
Staphylococcus lentus	Before milking	1	13									1	2
	After milking												
	Dry cows												
Staphylococcus pettenkoferi	Before milking												
	After milking												
	Dry cows							3	19			3	4

Continued

Table 5 (Continued). Distribution of 833 phenotypically different non-*aureus* staphylococci (NAS) on teat apices originating from lactating cows (n = 15), sampled before and after milking, and from dry cows (n = 18) in 5 commercial dairy herds

		Herd (H)											
		H1		H2		H3		H4		H5		Total	
NAS species	Teat apex strata	N[1]	%[2]	N	%	N	%	N	%	N	%	N	%
Staphylococcus rostri	Before milking												
	After milking												
	Dry cows									1	8	1	1
Staphylococcus saprophyticus	Before milking					1	6					1	2
	After milking												
	Dry cows												
Staphylococcus sciuri	Before milking	9	88	6	50							15	22
	After milking	3	25	3	25							6	8
	Dry cows	14	44					1	6			15	11
Staphylococcus simulans	Before milking												
	After milking												
	Dry cows					1	13	7	25			8	7
Staphylococcus succinus	Before milking			7	42					1	6	8	10
	After milking	1	13	3	25							4	7
	Dry cows	8	31									8	7
Staphylococcus vitulinus	Before milking	4	38					1	13			5	7
	After milking	2	13			1	6					3	3
	Dry cows	1	6									1	1
Staphylococcus xylosus	Before milking	2	13	1	67							12	15
	After milking	1	13	6	50					1	6	8	13
	Dry cows	19	63	1	5			5	19			25	19

[1]Number of phenotypically different (per sample) NAS isolates per teat apex stratum and herd.

[2]Percentage of sampled teat apices positive for a NAS species per teat apex strata and herd (teat apices can harbor more than 1 NAS species).

Table 6. Distribution of samples positive for only non-*aureus* staphylococci (NAS) in the 3 quarter milk strata: healthy quarters, quarters with subclinical, and clinical signs in commercial dairy herds and the presence of selected virulence and methicillin-resistance genes in NAS originating from healthy quarters and quarters with clinical signs[1]

	Quarter milk						
	≤50,000 cells/mL		>50,000 cells/mL		Clinical signs		Total
NAS species	N[2]	%[3]	N	%	N	%	N
Staphylococcus chromogenes	2	8	18	72	5[4]	20	25
Staphylococcus haemolyticus			10	67	5[5,6]	33	15
Staphylococcus equorum	3[5]	43			4[5,6]	57	7
Staphylococcus hominis	2	40	1	20	2	40	5
Staphylococcus epidermidis			1	25	3[6,7]	75	4
Staphylococcus xylosus			4	100			4
Staphylococcus devriesei	3	100					3
Staphylococcus simulans					2[6]	100	2
Staphylococcus auricularis			1	100			1
Staphylococcus capitis	1[6]	100					1
Staphylococcus cohnii	1	100					1
Staphylococcus arlettae/*S. xylosus*	1	100					1
S. chromogenes/*S. epidermidis*			1	100			1
S. chromogenes/*S. equorum*[5]					1[5]	100	1
S. chromogenes/*S. haemolyticus*[6]					1[6]	100	1
S. chromogenes/*S. hominis*	1	100					1
S. chromogenes/*Staphylococcus sciuri*			1	100			1
S. devriesei/*S. haemolyticus*			1	100			1
S. devriesei/*S. hominis*			1	100			1
S. devriesei/*S. xylosus*			1	100			1
S. epidermidis/*S. haemolyticus*			1	100			1
S. haemolyticus/*S. hominis*			1	100			1
S. sciuri[6]/*S. xylosus*[5,6]					1[5,6]	100	1
Staphylococcus succinus[6,7]/*S. xylosus*[6]					1[6,7]	100	1
S. equorum[5]/*S. haemolyticus*[6]/*Staphylococcus lentus*[6]					1[5,6]	100	1
S. equorum[5]/*S. haemolyticus*[6]/*S. xylosus*[5]					1[5,6]	100	1
S. auricularis/*S. devriesei*/ *S. haemolyticus*/*S. hominis*			1	100			1
Total	14	17	43	51	27	32	84

[1]No other bacteria were grown from those quarter milk samples.

[2]Number of quarters positive for 1 or more NAS species.

[3]Percentage of (the combination of) NAS species, present in healthy quarter milk samples, samples from quarters with subclinical or with clinical signs.

[4]Samples harboring at least 1 NAS isolate positive for the virulence gene *cap5H*.

[5]Samples harboring at least 1 NAS isolate positive for the virulence gene *bap*.

[6]Samples harboring at least 1 NAS isolate positive for the methicillin-resistance gene *mecA*.

[7]Samples harboring at least 1 NAS isolate positive for the virulence gene *agrA*.

81,000 to 212,000 cells/mL of milk) and 177,000 cells/mL of milk (IQR from 78,000 to 238,000 cells/mL of milk), respectively. These milk samples had a higher LnqSCC than bacteriologically negative milk samples ($P = 0.000$ and $P = 0.001$, respectively) and samples harboring *Corynebacterium* spp. ($P = 0.008$ and $P = 0.049$, respectively), yet the LnqSCC was not different from the major pathogen–positive milk samples.

Distribution of Virulence Genes. Table 7 outlines the results of NAS isolates positive for the virulence genes *agrA*, *bap*, and *cap5H* and methicillin-resistance gene *mecA*. At least one of those 4 genes was found in 71% of the 43 NAS isolates originating from quarters with signs of clinical mastitis, whereas only 25% of the NAS isolates from quarters with a qSCC ≤50,000 cells/mL were harboring either *bap* or *mecA*. Other virulence genes and methicillin-resistance genes were either absent in all tested isolates (*cap8H*, *clfA*, *coa*, *fnbpA*, *fnbpb*, *hla*, *hlb*, *ica*, *mecC*, *nuc*, *ssl7*, and *vwb*) or present in all tested isolates (*eno*).

DISCUSSION

This study aimed at extending our knowledge of the species-specific NAS presence and distribution in bovine quarter milk, on teat apices, and in rectal feces. The presence of NAS in rectal feces, reported recently by our group for the first time (Wuytack et al., 2019), was confirmed. Actually, 28% of the cows, ranging from 6% to 55% per herd, were shedding NAS in the feces. Moreover, we were able to study the presence of NAS over the 3 habitats in detail, expanding our knowledge

on the ecology of this group of staphylococci. Nine species, namely *S. arlettae*, *S. auricularis*, *S. chromogenes*, *S. cohnii*, *S. devriesei*, *S. equorum*, *S. haemolyticus*, *S. hominis*, and *S. vitulinus*, were present in all habitats in the same herds. In addition, the virulence potential of NAS was evaluated by comparing their distribution in milk samples originating from healthy quarters and quarters with signs of clinical mastitis, and by determining the presence of selected virulence genes.

This is not the first species-specific study that focuses on the distribution of NAS in quarter milk and on teat apices (De Visscher et al., 2016a; Mahmmod et al., 2018), yet we also included teat apex samples from both lactating and dry cows and rectal feces as a potential NAS source. The same species from rectal feces were also recovered from either quarter milk or teat apices, or both, in the same herds. These NAS species were further subtyped with the use of the strain-typing technique, random amplification of polymorphic DNA (**RAPD**)-PCR. For some species, the same RAPD type was found in rectal feces, on teat apices, and in quarter milk originating from the same herd, indicating that fecal NAS are a potential cause of NAS IMI (Wuytack et al., 2020).

Non-*aureus* staphylococci were recovered from 33% of the quarter milk samples, including samples that also yielded other pathogens, which is comparable to other studies (Supré et al., 2011; Young et al., 2015; Condas et al., 2017b), but mixed growth samples with 2 different species were not included in the study of Condas et al. (2017b). In heifers, the reported prevalence of NAS in milk is even higher (Piepers et al., 2010; De Visscher et al., 2016a) and lower NAS prevalence in milk samples, ranging from 6% to 13%, has also been reported (Paradis et al., 2010; Piessens et al., 2011; Mørk et al., 2012). Still, the large variation in study criteria (number of colonies, pure versus mixed culture) and identification methods between studies hampers meaningful comparison. Systematic sampling of quarter milk, teat apices, and rectal feces was used because it was the more practical way on farm. Bias might have been introduced if the factor under study was related to the sampling interval (Dohoo et al., 2014), which we believe was not the case. The 5 most prevalent species in this study were *S. chromogenes*, *S. haemolyticus*, *S. equorum*, *S. hominis*, and *S. cohnii*. Other studies also found a high prevalence of *S. epidermidis*, *S. simulans*, and *S. xylosus* (Sampimon et al., 2009a; Condas et al., 2017b; Jenkins et al., 2019).

The prevalence of NAS-positive teat apices is high in our study. The teat apices of dry cows all harbored NAS (1–8 species), which is comparable to the findings of Adkins et al. (2018a). Of the lactating cows, 94% of the teat apices were positive for NAS (also 1–8 species), which is similar to what Braem et al. (2013) reported. Other studies reported a lower prevalence of NAS on teat apices, ranging from 59.3% to 87% (De Visscher et al., 2014, 2016b; Mahmmod et al., 2018). The NAS isolates belonged to 22 different species, with a clear variation in distribution among herds, a finding that has been observed by others (De Visscher et al., 2016b; Adkins et al., 2018a). The predominant species on teats in our study were *S. haemolyticus*, *S. equorum*, and *S. cohnii*. These findings are in alignment with other reports (Braem et al., 2013; De Visscher et al., 2014; Mahmmod et al., 2018), although *S. cohnii* was mostly present on one herd (De Visscher et al., 2014) and on the teat apices before the milking process (De Visscher

Table 7. Distribution of virulence genes *agrA*, *bap*, and *cap5H*, and the methicillin-resistance gene *mecA* in 59 non-*aureus* staphylococci (NAS) isolates originating from quarter milk from healthy quarters (n = 16) and quarters with clinical signs (n = 43)

NAS species	$N_{isolates}$	*agrA*	*bap*	*cap5H*	*mecA*	$N_{pos.\ isolates}$[1]	$\%_{pos.\ isolates}$[2]
SCC ≤50,000 cells/mL	16						
Staphylococcus capitis	1				1	1	100
Staphylococcus equorum	3		3			3	100
Subtotal			3		1	4	25
Clinical signs	43						
Staphylococcus chromogenes	8			2		2	25
Staphylococcus epidermidis	4	4			4	4	100
S. equorum	9		9		2	9	100
Staphylococcus haemolyticus	11		1		9	9	82
Staphylococcus lentus	1				1	1	100
Staphylococcus sciuri	1				1	1	100
Staphylococcus simulans	2				1	1	50
Staphylococcus succinus	1	1			1	1	100
Staphylococcus xylosus	3		2		2	3	100
Subtotal		5	12	2	21	31	72
Total	59	5	15	2	22	35	59

[1]Number of NAS isolates positive for *agrA*, *bap*, *cap5H*, or *mecA* (or a combination of these).

[2]Percentage of NAS isolates positive for *agrA*, *bap*, *cap5H*, or *mecA* (or a combination of these).

et al., 2016b). *Staphylococcus simulans* was only isolated from the teat apices of dry cows, which is corroborated by the results of Mahmmod et al. (2018). Other species that were only isolated from dry cows were *S. pettenkoferi*, *S. fleurettii*, and *S. rostri*. A notable difference in the number of NAS isolates before (n = 280) and after the milking process (n = 155) on the teat apices supports the findings of De Visscher et al. (2014), suggesting that NAS are washed away during the milking process or susceptible to the premilking disinfectant products (when used). Still, *S. epidermidis* was only isolated after the milking process. As *S. epidermidis* is regularly isolated from infected quarters, cross-infection via the milk machine liners could have been occurred. Milkers' skin is also known as an infection source for NAS IMI (Thorberg et al., 2006; Piessens et al., 2011; De Visscher et al., 2014).

At least one dairy cow in every herd was shedding NAS species in feces, where 28% of the rectal feces samples were NAS-positive. The latter was lower than what we reported earlier (52%; Wuytack et al., 2019), yet in that study we only sampled cows (repeatedly) from one herd. *Staphylococcus rostri* was only identified in quarter milk and on teat apices in the present study but not in rectal feces, whereas it was the most prevalent species in our previous study. This suggests a herd-specific distribution of NAS in feces exists as well, as is true for NAS from quarter milk and teat apices (Piessens et al., 2011; De Visscher et al., 2016b). The NAS species originating from rectal feces were also found both on teat apices and in quarter milk in the same herd, as aforementioned. Strain-typing NAS, present in the 3 habitats on the same herds, revealed that the same RAPD type was present in each of them for some species. These findings suggest that fecal NAS can infect the bovine mammary gland (Wuytack et al., 2020).

The occurrence of NAS in milk samples (10%) was similar to another study where NAS recovery in pure culture was studied, not taking into account samples yielding other pathogens (Paradis et al., 2010) but dissimilar to the study of Condas et al. (2017b), with a NAS prevalence of 26% in pure culture. The difference in study prevalence of NAS could be explained by the use of a semi-selective agar, MSA, in our study. This agar improves the differentiation of phenotypically different colonies from NAS (De Visscher et al., 2013) just as we observed in the current study, but we also observed better growth on MSA than on esculin blood agar of certain *Bacillus* spp. and *Corynebacterium* spp. We corroborated that *S. chromogenes*, *S. haemolyticus*, and *S. xylosus* are 3 of the 5 most prevalent species causing bovine IMI (Vanderhaeghen et al., 2014; Condas et al., 2017a), yet *S. equorum* and *S. hominis* completed the top 5 most isolated species. *Staphylococcus simulans* was isolated only 2 times in our study, unlike previous studies (Mørk et al., 2012; Nyman et al., 2018).

The NAS as a group was found in 12% of the quarter milk samples originating from clinical mastitis cases, comparable to other studies that report a NAS prevalence in clinical mastitis cases of 5% ranging up to 20% (Persson Waller et al., 2011; Verbeke et al., 2014; Levison et al., 2016). The NAS-positive samples from clinical mastitis cases range from 0% in herd 8 to 25% in herd 2. This difference between herds might be explained by the variation in NAS species and their virulence potential. Herd-specific factors (e.g., housing conditions, clipping of the udder, and milking protocol) also influence species-specific prevalence as well as seasonal effects, although more research is required to identify species-specific risk factors (De Visscher et al., 2016a, 2017; Dolder et al., 2017).

Intramammary infection with NAS as a group caused a moderate elevation in qSCC. The elevation in qSCC (109,000 cells/mL of milk) is situated between the geometric mean of 70,000 cells/mL found by Condas et al. (2017a) and 137,000 cells/mL of milk reported by Supré et al. (2011). This variation in qSCC could be explained by the difference in NAS distribution because the level of elevation in SCC is species specific (Fry et al., 2014; Nyman et al., 2018). The NAS-positive milk samples (all species combined) had a significantly higher LnqSCC than the milk samples yielding no growth and *Corynebacterium* spp. The qSCC was not different from quarter milk with major mastitis pathogens, suggesting that NAS as a group have some effect on udder health. Still, only 6 quarters were major pathogen positive, reducing the power of finding true differences. Furthermore, some NAS species tend to be more relevant for the udder health because samples harboring NAS species other than *S. chromogenes* or *S. haemolyticus* only did not have a significantly higher LnqSCC than samples yielding no growth or *Corynebacterium* spp. (data not shown). This finding is in alignment with other studies (De Visscher et al., 2015; Condas et al., 2017a; Adkins et al., 2018b).

Virulence genes *agrA*, *bap*, and *cap5H* were found in 44% of the NAS isolates originating from quarters with clinical signs, which is in contrast to healthy quarters of which only 19% were harboring *bap*. Naushad et al. (2019) investigated the profile of 191 virulence factors in NAS, based on whole-genome sequencing, and found distinctive patterns of associations for low qSCC (defined as <150,000 cells/mL in that study) and clinical mastitis isolates. This finding is in alignment with our results, although we only tested a limited number of virulence genes (n = 15). Despite the relatively high number of phenotypically identified *Staphylococcus* spp.

isolates (n = 1,676) in our study, only a limited number of molecularly identified NAS (n = 134) were present in quarter milk samples harboring NAS only. In addition to the limited number of NAS isolates, they belonged to several NAS species (n = 15) divided over the 3 different milk strata, leaving an insufficient number for species-specific statistical analysis regarding the virulence potential.

All isolates tested positive for the biofilm-related gene, *eno*, a finding that is comparable to other studies that also reported a prevalence as high as 88% and 92.6%, respectively (Darwish and Asfour, 2013; Srednik et al., 2017). In our study, the prevalence of the *mecA* gene in NAS isolates from clinical mastitis cases was 49%, situated between the *mecA* prevalence of 29% described by Khazandi et al. (2018) and the 95% reported by Mahato et al. (2017). Still, Khazandi et al. (2018) tested only oxacillin-resistant isolates instead of all NAS isolates originating from clinical mastitis cases, as in our study, and might have missed oxacillin-susceptible *mecA*-positive staphylococci (Mahato et al., 2017). It is remarkable that *mecA* was found in 49% of the NAS isolates originating from clinical mastitis and only in one isolate (6%) originating from healthy quarters. Although the presence of certain antimicrobial genes may be associated with reduced fitness of the strain, the methicillin-resistance gene, *mecA* in bovine NAS isolates might be linked with virulence genes or pathogenicity islands, supposedly both present on the mobile element, SCC*mec*. The presence of virulence genes and the methicillin-resistance gene, *mecA*, dovetails nicely with the prevalence of NAS in the opposite strata of the quarter milk because NAS species were recovered 3 times as frequently in samples from clinical mastitis cases compared with milk samples from healthy quarters, suggesting that some NAS isolates have the ability to cause (mild) clinical mastitis.

CONCLUSIONS

This study reaffirms the presence of NAS in rectal feces from dairy cows in every sampled herd, with a range of 6% to 55% of the cows shedding NAS. Nine species, namely *S. arlettae*, *S. auricularis*, *S. chromogenes*, *S. cohnii*, *S. devriesei*, *S. equorum*, *S. haemolyticus*, *S. hominis*, and *S. vitulinus*, were present in quarter milk, teat apices, and rectal feces in the same herds. Further research should investigate the potential of fecal NAS as a mammary gland infection source or NAS teat apex colonization, or both. Furthermore, our findings suggest that NAS have the ability to cause mild clinical mastitis because the occurrence of those NAS, in milk samples harboring NAS only, was larger in those originating from clinical mastitis (12%) than the occurrence in milk samples from healthy quarters (4%). Studying the distribution of virulence genes in a selection of NAS isolates from 2 contrasting milk sample strata strengthened this finding.

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